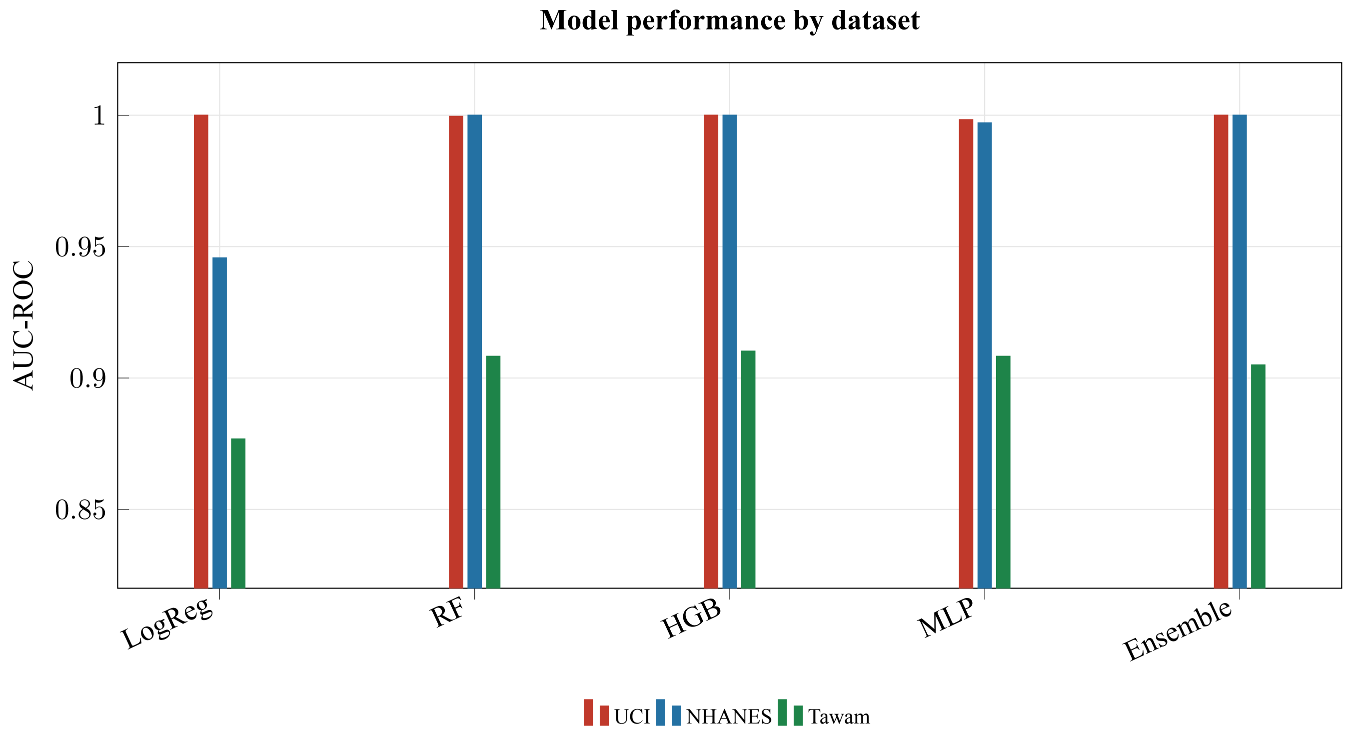
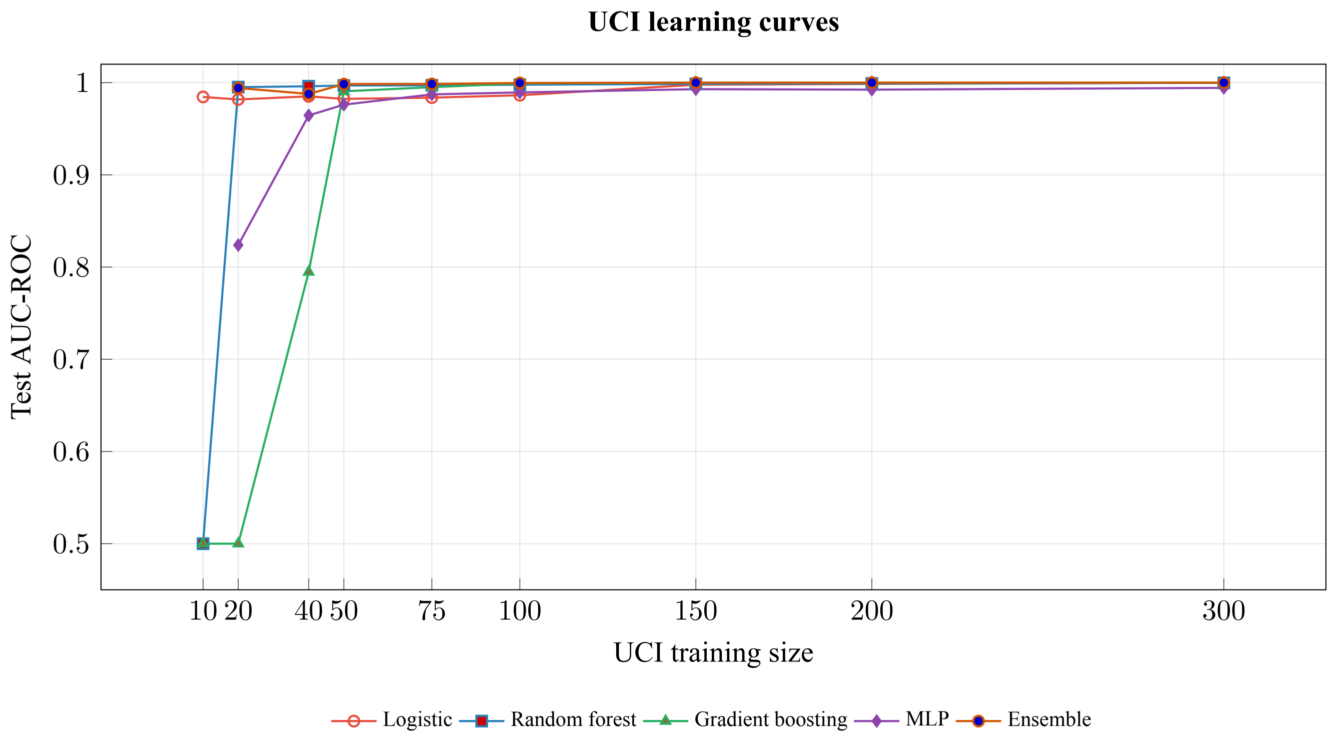
Supplementary Information: Figures

When Perfect Is Wrong: Exposing the AUC ≈ 1.0 Illusion in Chronic Kidney Disease Machine Learning Benchmarks

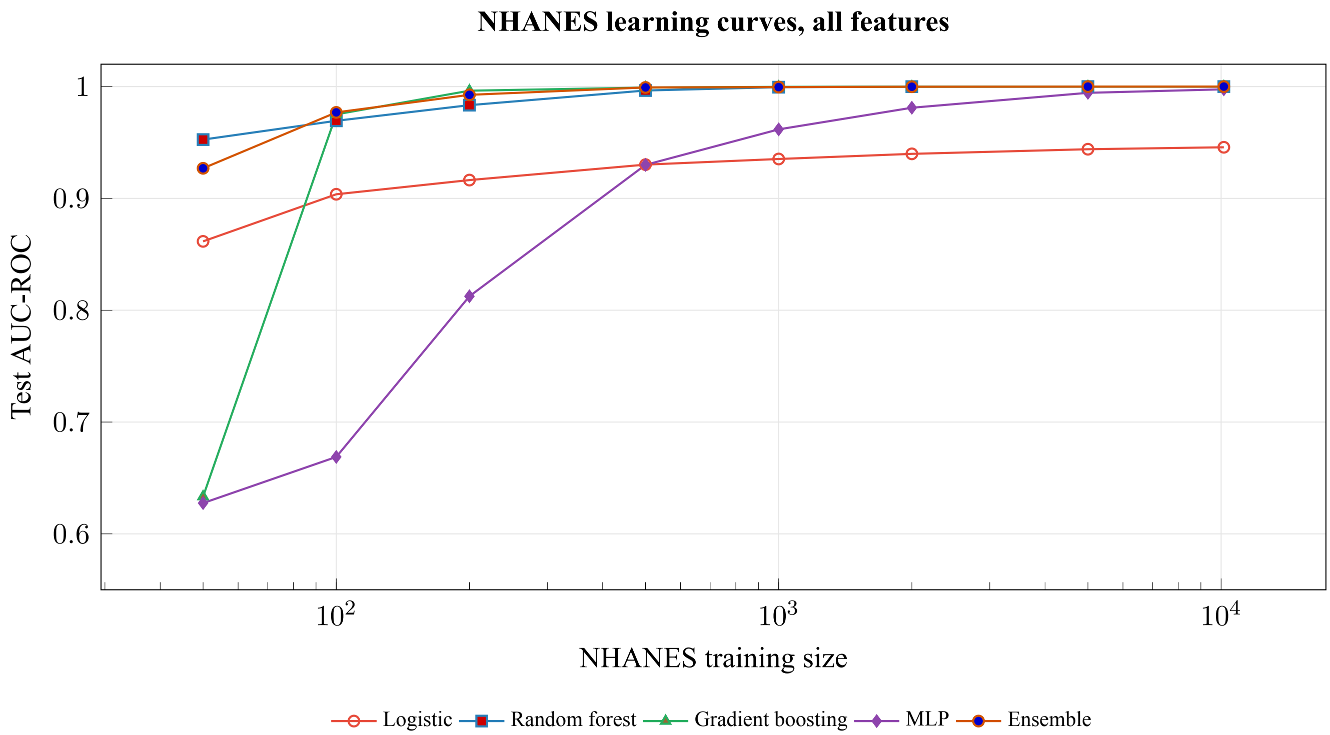
This supplementary document contains the figures referenced by the online copy of the manuscript. Figures are reproduced from the LaTeX-native manuscript render using the regenerated CSV outputs.



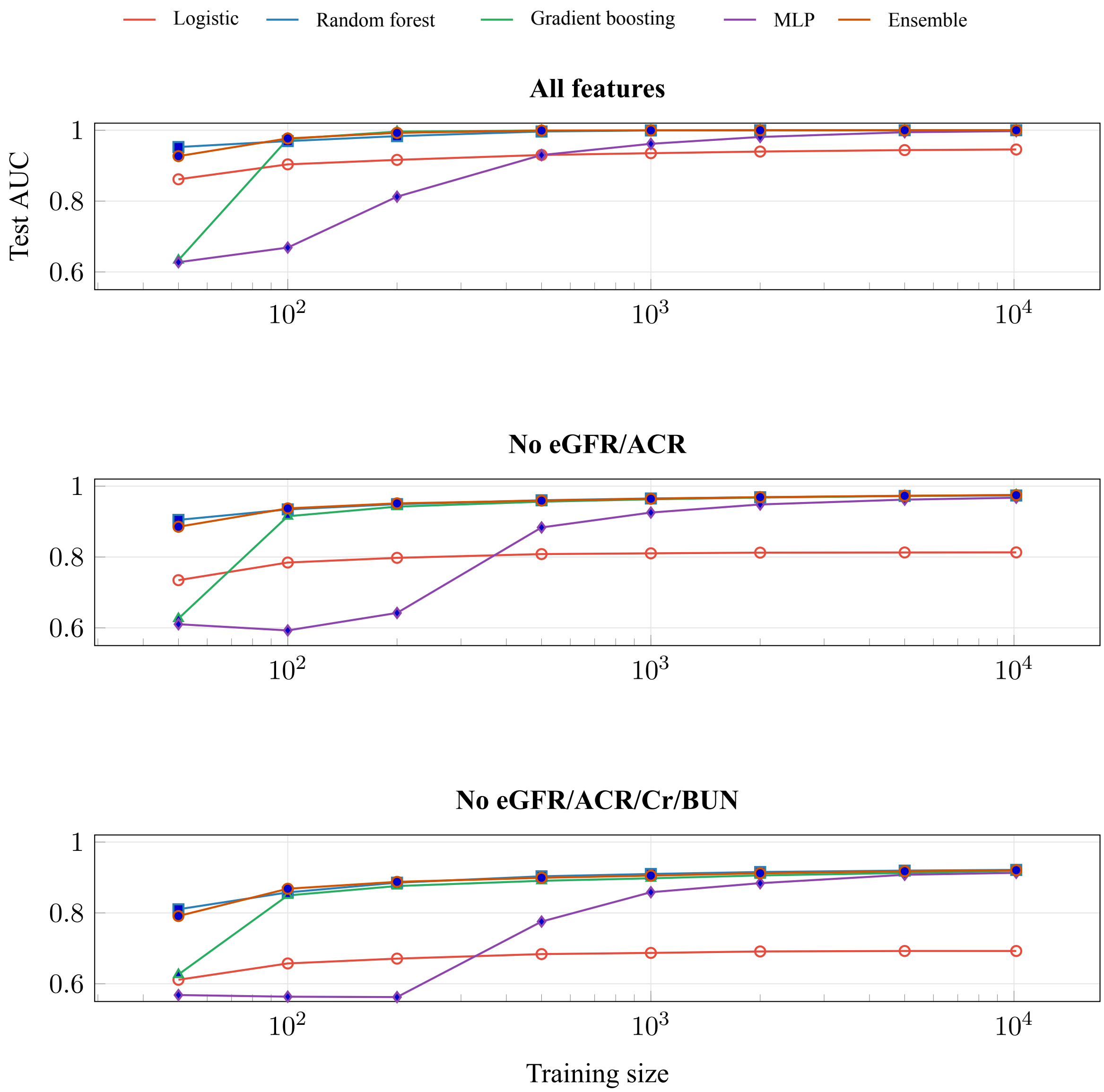
Supplementary Figure S1: UCI is saturated across model families, NHANES with all features is also near-ceiling because of label-feature overlap, and Tawam remains lower and more discriminating among algorithms. The y-axis is intentionally truncated at 0.82 to make small AUC differences visible.



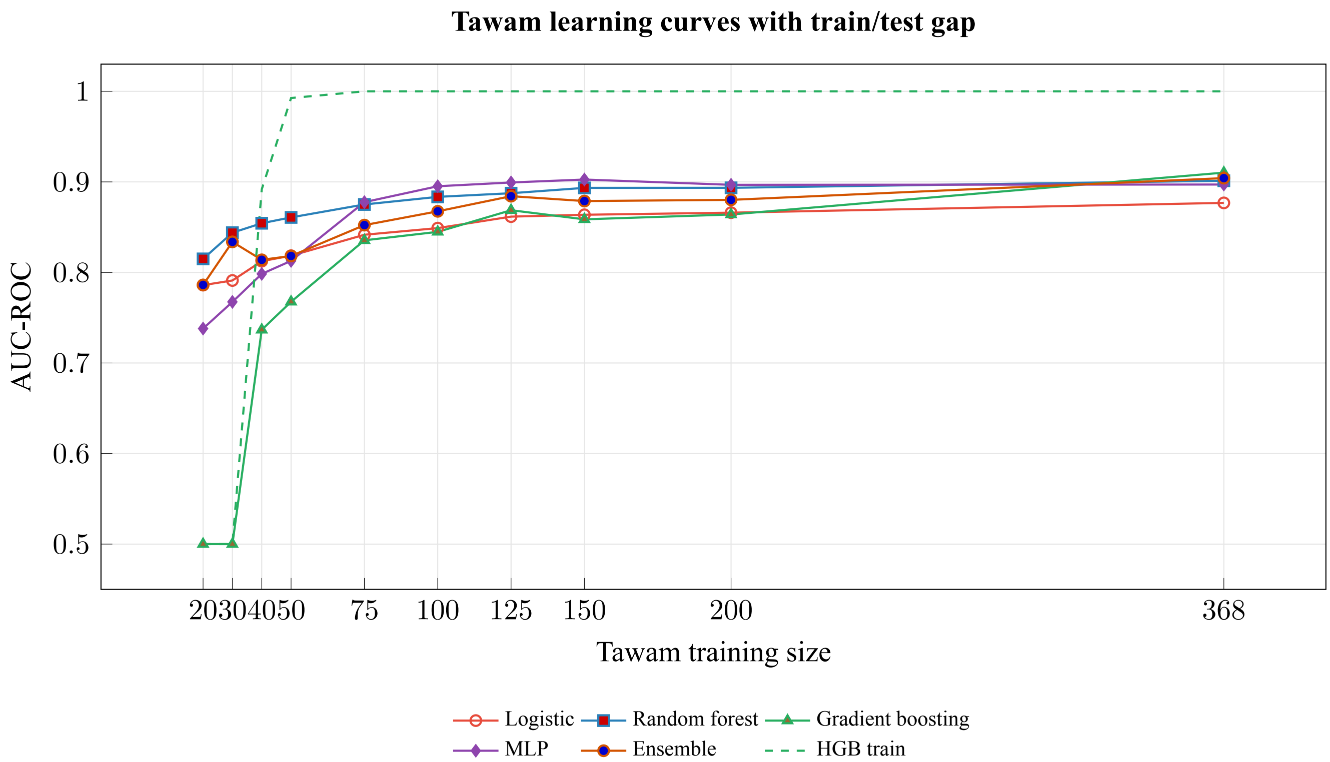
Supplementary Figure S2: UCI performance reaches near-perfect AUC with small stratified subsamples, supporting the ceiling-effect interpretation.



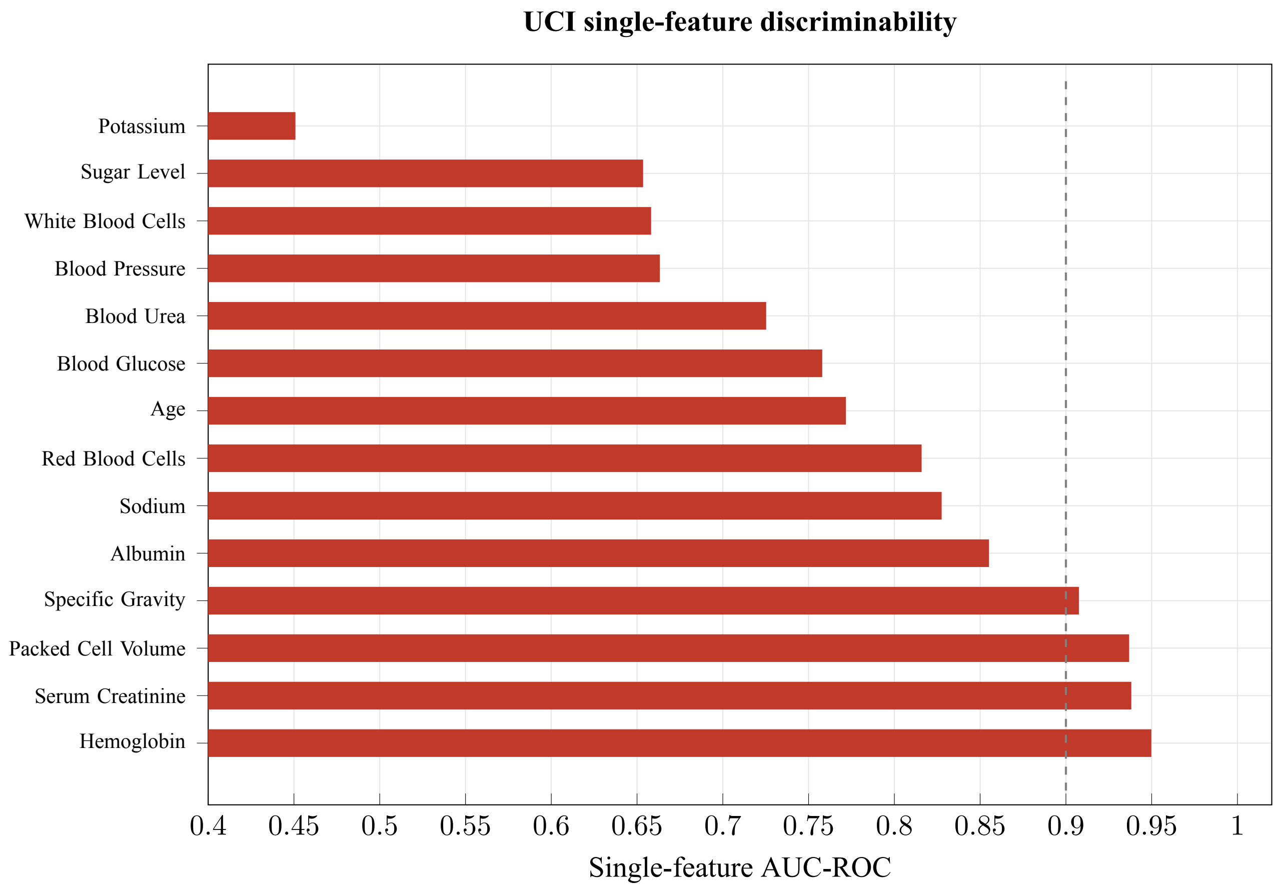
Supplementary Figure S3: With eGFR and ACR included, NHANES approaches a ceiling because the label definition is partly inside the feature set under KDIGO-style eGFR/albuminuria criteria [1].



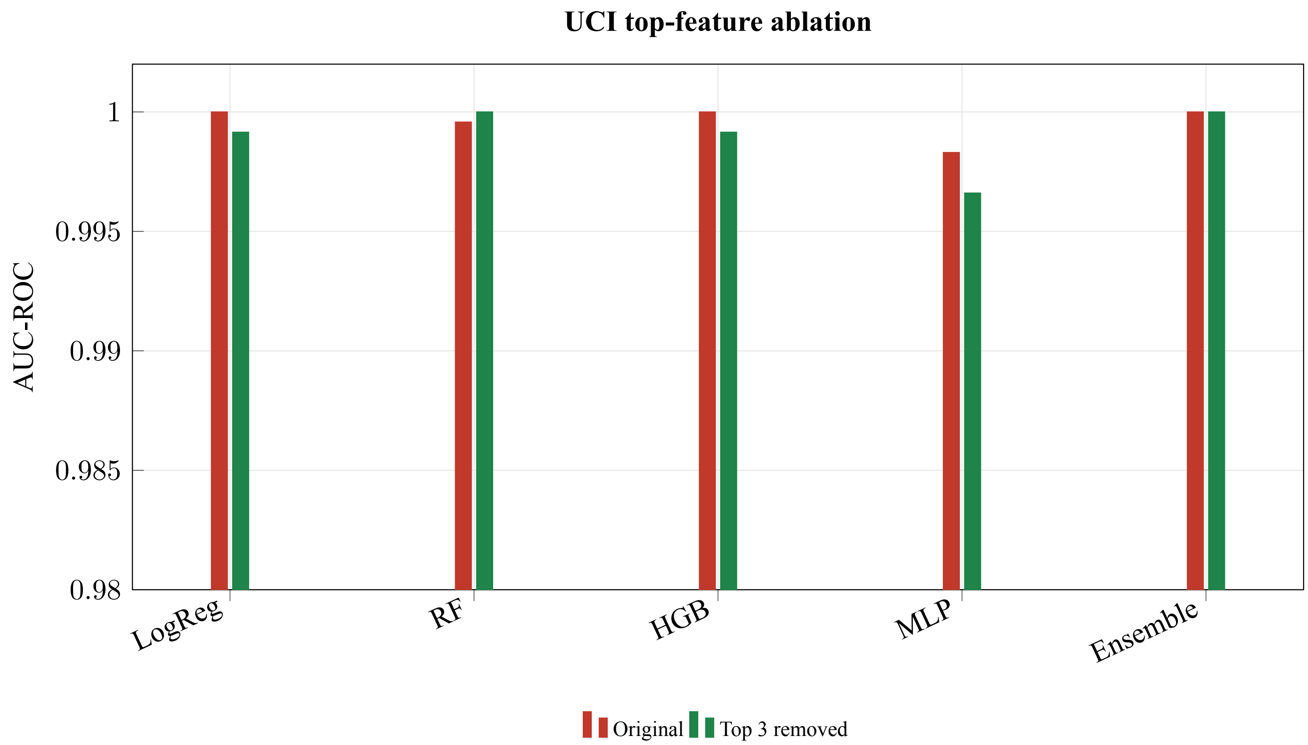
Supplementary Figure S4: Removing label-defining and closely related laboratory variables exposes a substantially harder NHANES prediction task, especially for logistic regression.



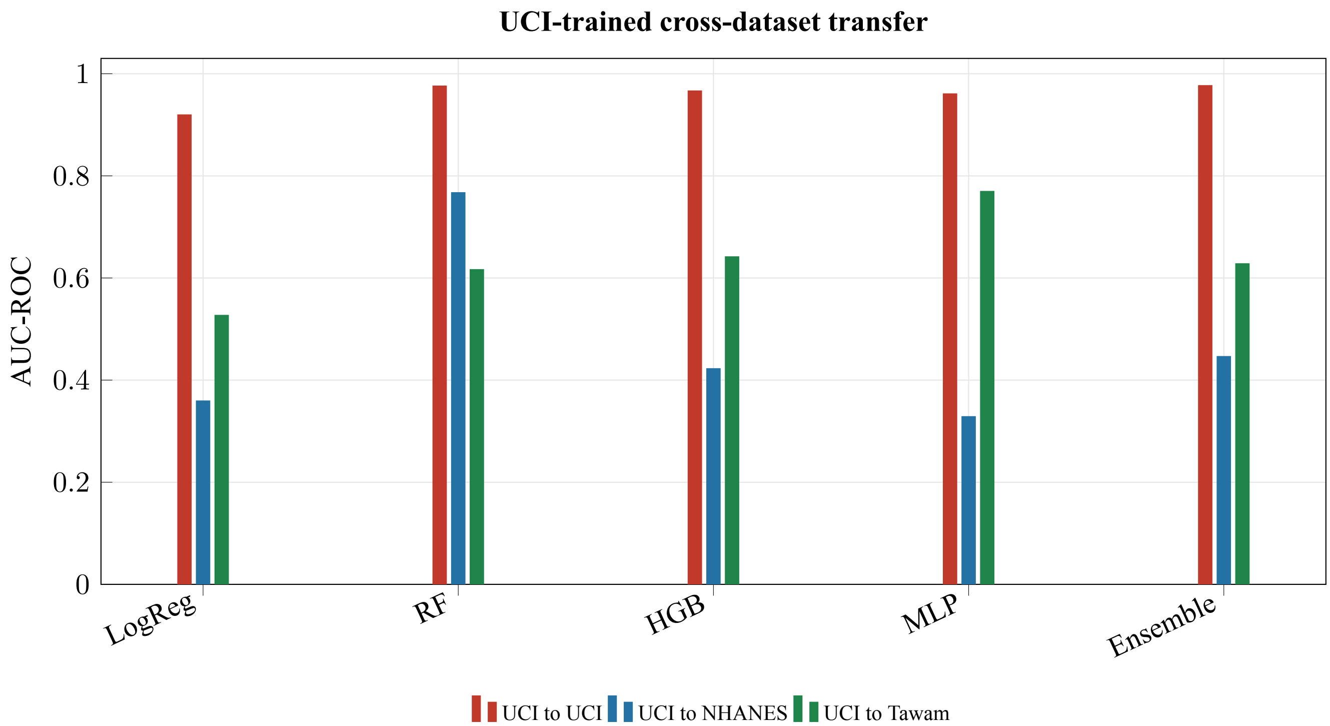
Supplementary Figure S5: Tawam test AUC improves with additional data. The dashed gradient-boosting training curve shows the monitored overfitting gap.



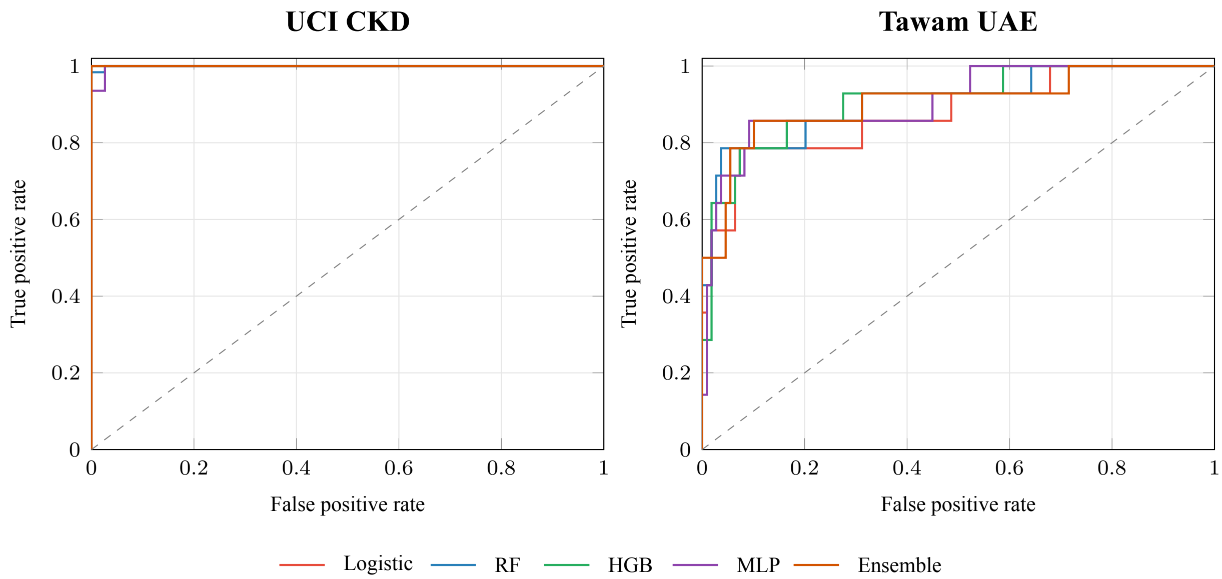
Supplementary Figure S6: Several individual UCI features approach or exceed AUC 0.90 on their own.



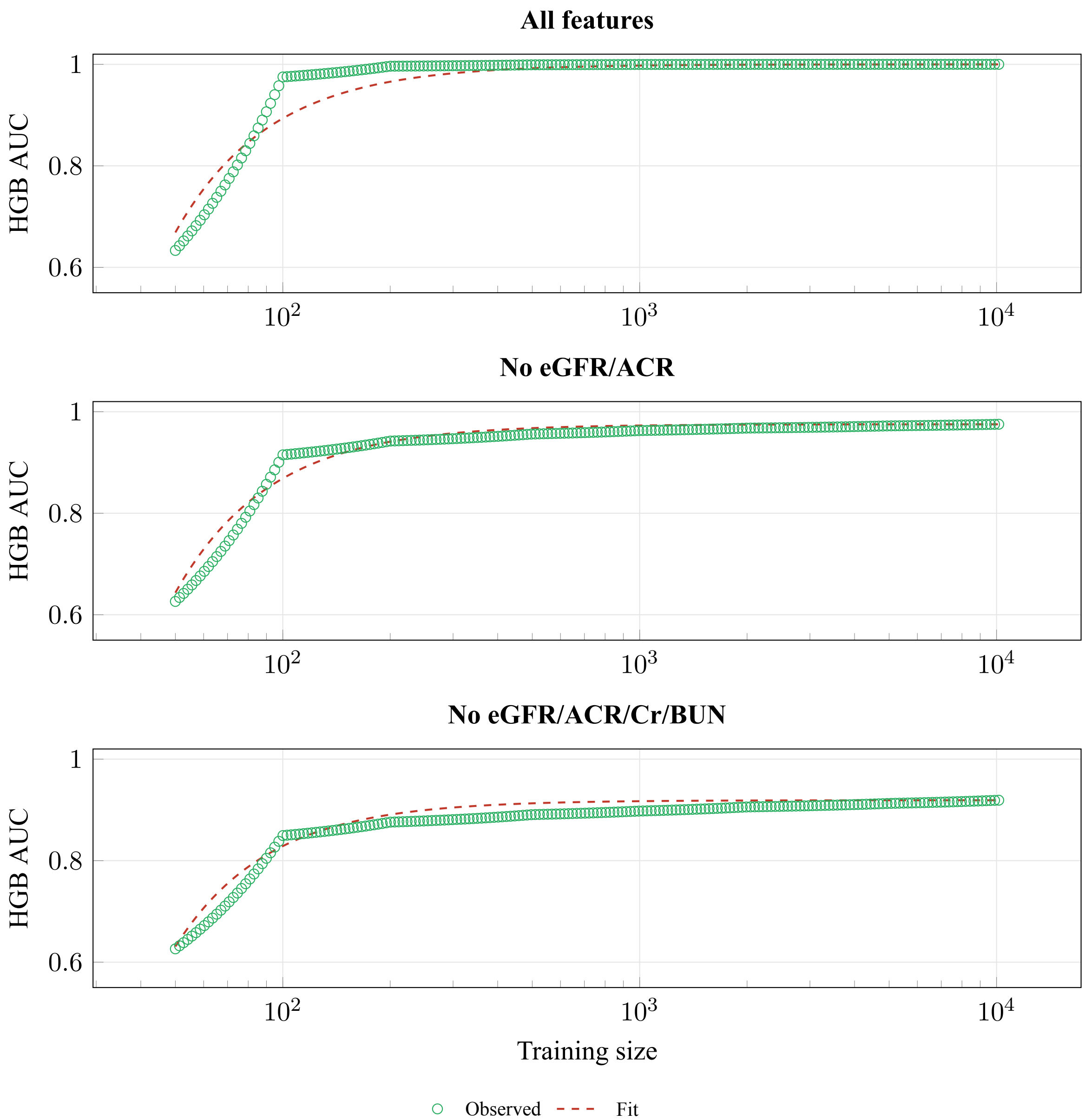
Supplementary Figure S7: Top-feature removal barely lowers UCI performance. The 0.98-truncated y-axis makes the near-ceiling range visible.



Supplementary Figure S8: UCI-trained models do not reliably transfer to NHANES or Tawam even when the feature space is restricted to shared clinical variables.



Supplementary Figure S9: ROC curves collapse near the upper-left corner on UCI, while Tawam preserves more visible separation.



Supplementary Figure S10: Power-law fits summarize NHANES learning behavior under the three feature definitions. The ceiling parameter is descriptive and constrained by AUC's upper bound.

References

[1] KDIGO CKD Work Group. KDIGO 2024 clinical practice guideline for the evaluation and management of chronic kidney disease. *Kidney International.* Vol. 105, No. 4S, pg. S117–S314, 2024, DOI: `https://doi.org/10.1016/j.kint.2023.10.018`.